

Gimbal Mount Assembly

Justification File

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1. Introduction

1.1. Scope

This document presents the Design Justification File for the Gimbal Mount Assembly (GMA). Its purpose is to explain the rationale behind the selected design and demonstrate that it meets the baseline requirements specified in [RD02]. It lists and describes the justification for:

- Functional Design
- Thermal Design
- Mechanical Design

1.2. Reference

[RD1] A.Lukanowski - Gimbal Mount Assembly - Requirement Consolidation / Version 0 - Luebeck 2026

[RD02] Gimbal Mount Assembly - Requirement Consolidation / Version 001

[RD03] SKF Spherical Plain Bearings and Rod Ends / 2013

[RD04] Bearings, Control System Components, and Associated Hardware Used in the Design and Construction of Aerospace Mechanical Systems and Subsystems / MIL-HDBK-5199 / 1997

[RD05] RBC Aerospace FibriloidCR Series - Cryogenic Rated Plain Bearings: Sphericals, Rod Ends and Journals / 2024

[RD06] SKF Aerospace Solutions / 2020

Functional Design

Chosen Tolerances and fits

Figure 1 shows a cross-sectional view in which all radial interfaces are color-coded. **Table 1** summarizes the dimensional limits and resulting diametral fit range for each mating outside-diameter/inside-diameter interface. Positive fit values indicate clearance, while negative values indicate interference.

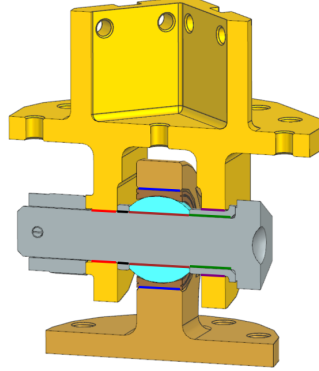


Figure 1: GMA - Radial tolerances color-coded

Table 1: Radial tolerances and fits

Mating parts	Shaft (mm)	Hole (mm)	Resulting Fit (mm)
Bearing & Lug	30.1498–30.1625	30.1570–30.1700	-0.0055 to +0.0202
Bolt & Bearing	15.8369–15.8496	15.8623–15.8750	+0.0127 to +0.0381
Bolt & Clevis	15.8369–15.8496	15.8623–15.8750	+0.0127 to +0.0381
Bolt & Bushing	15.8369–15.8496	15.8623–15.8750	+0.0127 to +0.0381
Bolt & Spacer	15.8369–15.8496	15.8750–15.902	+0.0254 to +0.0651
Bushing & Clevis	19.9800–19.9930	20.0000–20.0210	+0.0070 to +0.0410

The selection of the commercial off-the-shelf (COTS) MS14103-10 spherical bearing and NAS6710DU29 bolt constrains the available bearing and bolt-shank dimensions. The dimensions of the corresponding mating bores were selected to provide the required assembly clearances. The resulting bolt-to-bore clearances are also consistent with established bearing-supplier recommendations, including those provided by *SKF* [RD03].

To simplify assembly and inspection and to minimize the precautions required during integration, a common close-clearance fit is used for most bolt-to-bore interfaces. The exception is the transition fit between the GMA lug-head bore and the spherical-bearing outer ring. During bolt tightening, the clearance at the bolt interfaces allows the components to align and seat axially without radial binding.

The only transition fit in the assembly is between the GMA Lug Head bore and the Spherical Bearing outer ring. This fit intentionally differs from the generic recommendation in the Defense Handbook [RD04], which specifies a nominal 0.001-inch clearance fit.

In contrast to the generic recommendation, *SKF* [RD03] distinguishes between housing fits according to the housing material and the expected loading conditions. For steel/PTFE-fabric spherical bearings subjected to heavier loads or shock loading, *SKF* recommends a tighter housing fit. Applying the recommended *K7*-housing-bore tolerance to the outside-diameter limits of the *MS14103-10* Spherical Bearing results in a calculated fit range of **-0.0180 to +0.0197 mm**.

Other bearing suppliers, including RBC Aerospace [RD05] and NHBB [reference required], recommend a transition fit of approximately -0.005 to +0.020 mm for self-lubricating spherical bearings. The selected GMA dimensions produce a fit range of -0.0055 to +0.0202 mm, which closely matches this recommendation. This fit was selected because the GMA is expected to experience high oscillatory loads during steady-state operation and transient shock loads during ignition, shutdown, and combustion-instability events.

In the event of a interference of **-0.0055 mm** between Spherical Bearing and Lug, the clearance fit is expected to remain between the Bolt and Bearing. As a conservative screening calculation, assuming that the full outer-ring

diametral compression were transferred directly to the bearing bore would leave 0.0072 mm (0.0127 mm - 0.0055 mm) of diametral bolt-to-bearing clearance.

It should also be noted that a tighter housing fit can alter the installed bearing clearance and decrease the bearing friction coefficient at a given temperature, as described qualitatively for self-lubricating spherical bearings by *SKF* [RD06].

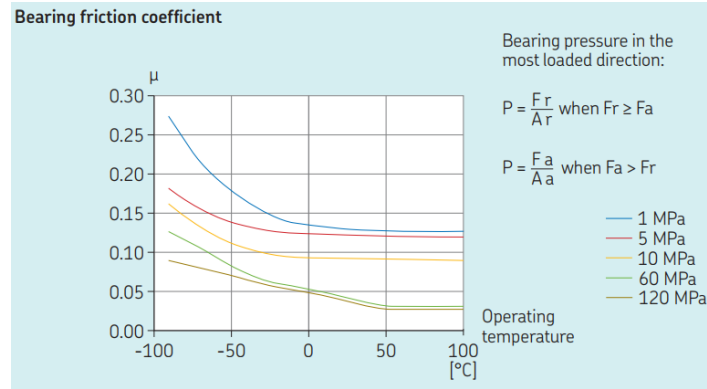


Figure 2: Load and temperature dependend evolution of bearing friction coefficient [RD06]

Tolerance stack-up analysis

A tolerance stack-up analysis is conducted with the objective to keep tolerances as tight as possible to maintain the design compact, reduce the roll-error (pivot on x-axis) and comply the geometrical constraints given by COTS parts utilized. **Figure 3** shows four cases of tolerance stack-up analysis performed to estimate the minimum and maximum clearance under worst-case considerations (absolute extreme tolerance limits)

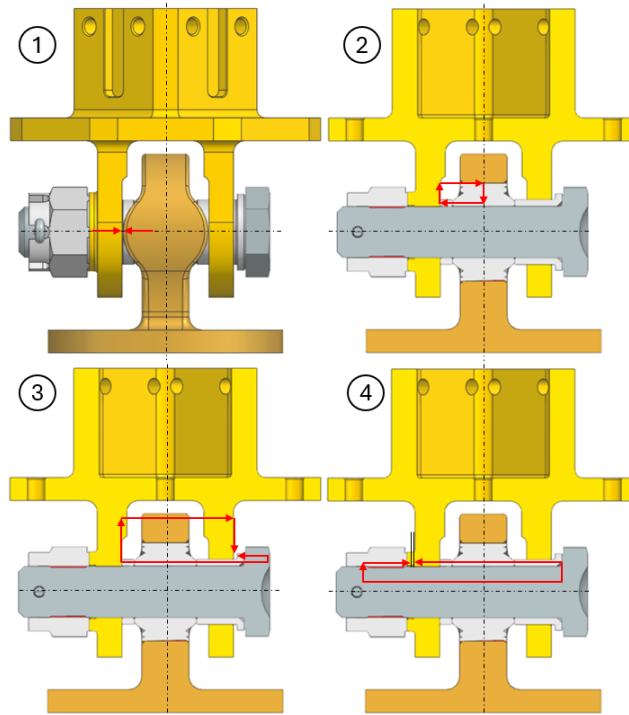


Figure 3: GMA tolerance stack-up

Table 2 presents the related values calculated. The detailed

Table 2: Worst-Case Tolerance Analysis

Stack-up	Nominal value (mm)	Max. value (mm)	Min. Value (mm)
1	0.3	0.6	0.1
2	0	0.0854	0.015
3	1.225	1.425	0.904
4	0.699	1.055	0.395

Preload of NAS-bolt

Thermal Analysis

Generic Boundary Conditions - Load assumptions - Convection, Radiation, Conduction

Mechanical Analysis

Bearing Lifetime

- Show first how size was determined @MIL-HDBK-1599 - 201.143
- Show computed lifetime according SKF
- Discuss results ## Bolt, Lug, Clevis
- Preliminary assesment Ro/Ma
- Show simplified assumptions and analytical result
- Show numerical result ## Static Structural
 - ## Preliminary Analysis ### Bolt, Lug, Clevis

Spherical Bearing: Liner Stiffness

GMA Static-Structural Analysis

- 0° with Thrust load and Acceleratoin
- Worst case angles (pitch/yaw)
- Discuss results
 - Modelling of liner (boded)

Conclusion

Annex

Tolerance Stack Analyis

Boundary Conditions Thermal Analysis

Boundary Conditions Static Structural Analysis

Tolerance Stack Analyis

3. Acronym List

The acronyms used in this document are listed below.

Acronym	Definition
COTS	Commercial Off-The-Shelf
GMA	Gimbal Mount Assembly